

**2** Plants and animals use chemicals to co-ordinate metabolic activities and development.

(a) (i) Which is a function of gibberellins in plants?

(1)

- ☐ **A** hydrolyse starch to release glucose
- ☐ **B** inhibit elongation of cells
- ☐ **C** regulate the translation of DNA
- ☐ **D** regulate the transcription of genes

(ii) Which is produced by an animal in a 'fight or flight' response?

(1)

- ☐ **A** adenine
- ☐ **B** adrenaline
- ☐ **C** amylopectin
- ☐ **D** antihypertensive

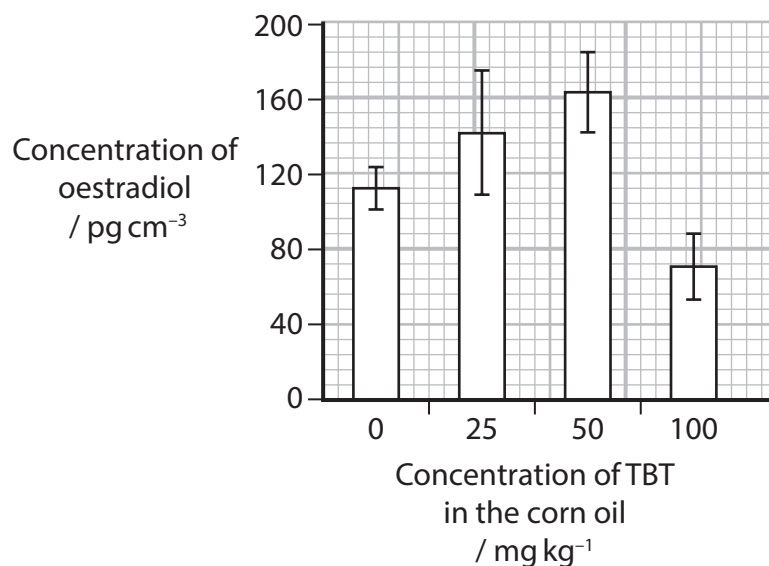
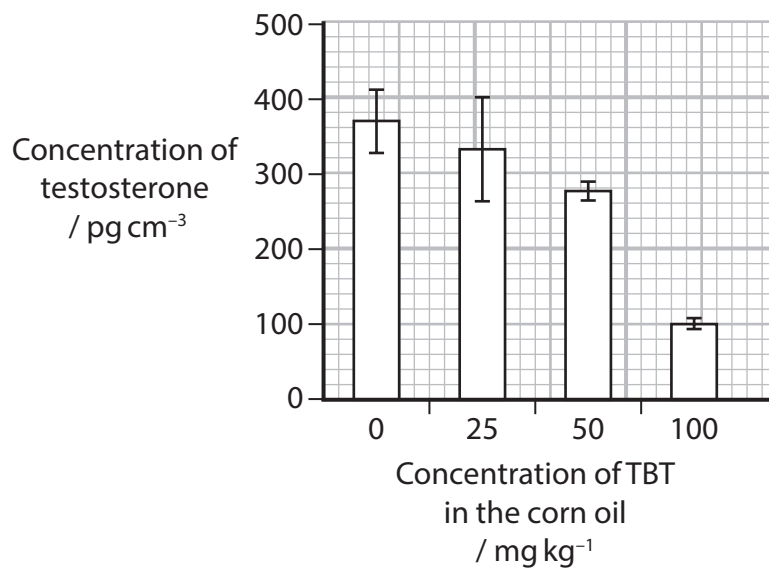
(b) Tributyltin (TBT) is a commonly used chemical in boat building.

The effect of TBT on the blood levels of two steroid hormones, testosterone and oestradiol, was investigated.

Groups of male mice were fed corn oil alone, or corn oil containing different concentrations of TBT.

The concentrations of testosterone and oestradiol in the blood of the mice was measured.

The results are shown in the graphs.



(i) Describe **three** conclusions that can be drawn from this investigation.

(3)

[illegible]

(ii) Describe how steroid hormones regulate gene transcription.

(3)

**(Total for Question 2 = 8 marks)**